B3.2 Transport (Animal)

Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/sl-b3-2-animal.html

Written for this site from the course textbook and class slides. Use the page numbers to read the book's fuller account. Water transport in plants (B3.2.7–10) is in Plant Systems. Image credits are on the last slide of the slides for this subtopic.

Exchange

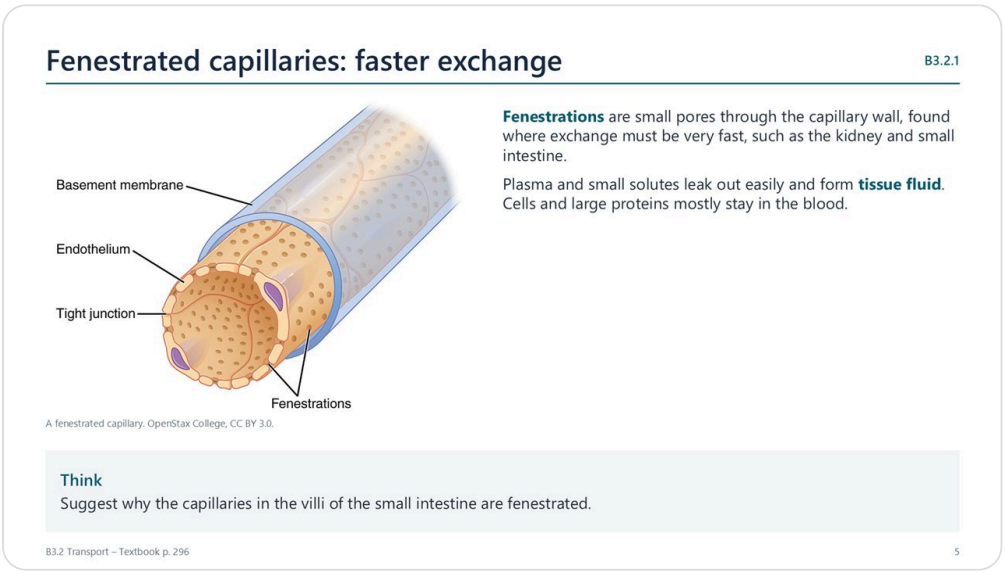
B3.2.1 How capillaries are adapted for exchange

p. 296

Capillaries are where materials pass between blood and tissue fluid. Each feature helps:

- **Branching** gives a huge surface area, and no cell is far from a capillary.
- **Narrow diameter**, about the width of one red blood cell, puts those cells right next to the wall.
- **Thin wall**, a single layer of endothelial cells, makes the diffusion distance short.
- **Fenestrations**, or small pores, occur in some capillaries where exchange has to be especially rapid, such as in the kidney and small intestine. Some plasma escapes to form tissue fluid.

Low pressure and slow flow also give more time for exchange.



Arteries and veins

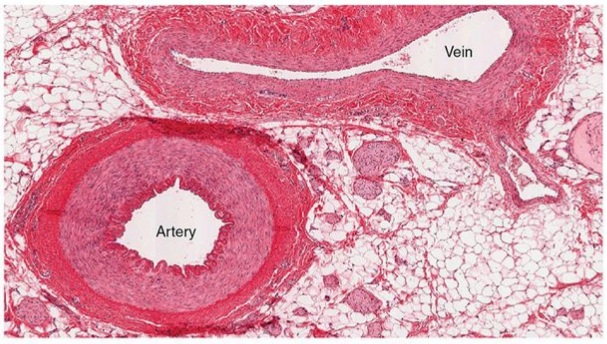
B3.2.2 The structure of arteries and of veins

pp. 296–297

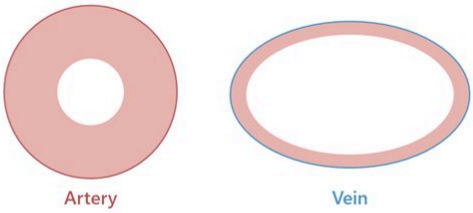
Both have three layers around a hollow **lumen**: the **tunica intima** (smooth endothelium), the **tunica media** (smooth muscle and elastic fibres) and the **tunica externa** (connective tissue with collagen).

Identifying vessels in a micrograph

B3.2.2



Light micrograph: vein (above) and artery (below). OpenStax College, CC BY 3.0.



Compare the **wall** with the **lumen**. Artery: thick wall, narrow lumen, round. Vein: thin wall, wide lumen, often squashed.

Common mistake
Do not judge by overall size. Use the wall thickness relative to the lumen.

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In a micrograph, compare the wall with the lumen. An **artery** has a thick wall and a narrow lumen and tends to look round. A **vein** has a thin wall and a wide lumen and often looks squashed.

Watch out: size alone does not tell you which is which. Use wall thickness relative to lumen diameter.

B3.2.3 How arteries are adapted to carry blood away from the heart

pp. 297–298

Each heartbeat sends a surge of blood into the arteries at high pressure, reaching about 120 mm Hg in the aorta. The wall has two jobs:

- **Withstand the pressure:** a thick wall of muscle, with collagen, prevents the artery from rupturing.
- **Maintain the pressure:** elastic fibres stretch as the surge arrives, storing energy, and then recoil between beats, pushing the blood on. Contracting smooth muscle also narrows the lumen.

Further from the heart, arteries have less elastic tissue and more smooth muscle. Arterioles use this muscle to control how much blood reaches each capillary bed.

B3.2.4 Measuring pulse rate

pp. 298–299

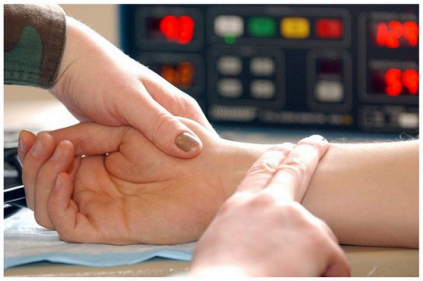
Each wave of blood from the ventricles stretches the arteries, and you can feel this where an artery runs near the skin and over bone: the **radial** artery above the wrist, or the **carotid** artery in the neck.

Press gently with two fingertips. Count for a full minute, or count for 30 seconds and double it. Do not use your thumb, as it has its own pulse.

Digital devices such as fitness watches shine an LED through the skin and detect changes in blood flow. They record continuously but can vary in accuracy.

Taking a pulse

B3.2.4



Feeling the radial pulse. Jeanette Copeland, US Air Force, public domain.

Each beat sends a **pressure wave** along the arteries. Feel it where an artery lies near the skin over a bone.

Radial: wrist, thumb side. **Carotid**: side of the neck. Count for 60 s, or 30 s and double.

Common mistake

Use your fingertips, not your thumb. The thumb has a pulse of its own.

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B3.2.5 How veins are adapted to return blood to the heart

p. 299

By the time blood reaches a vein it has lost nearly all its pressure, at about 5 mm Hg, and has no pulse. Backflow is a real risk, so veins rely on two features:

- **Valves:** blood moving to the heart pushes the flaps aside. If it flows backwards, it fills the pockets and the flaps close.
- **A thin, flexible wall:** contracting skeletal muscles can squeeze the vein, pushing blood along it.

The wide lumen also reduces friction. This is why walking helps blood to return from the legs.

When arteries become blocked

B3.2.6 What blocks coronary arteries and what follows

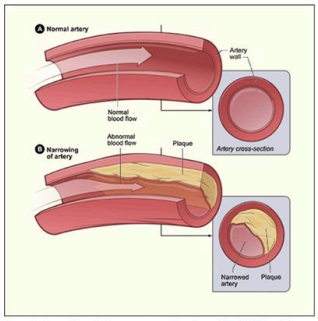
pp. 300–302

Coronary arteries branch from the aorta and supply heart muscle. **Occlusion** means that a vessel is blocked or closed.

1. Fatty deposits, or **atheroma**, form under the endothelium.
2. The lumen narrows and blood pressure rises, which damages the wall further.
3. Fibrous repair tissue makes the wall less elastic.
4. The lining breaks down, forming a **plaque**.
5. A ruptured plaque triggers clotting, and the clot is a **thrombus**. If it breaks free it is an **embolus**.

How arteries become blocked

B3.2.6



Normal artery (A) and artery narrowed by plaque (B). NHLBI, public domain.

Fatty **atheroma** forms under the endothelium

↓

Lumen narrows; wall hardens

↓

Plaque ruptures

↓

A clot forms: a **thrombus**

↓

Artery blocked: occlusion

Embolus
A clot that breaks free and lodges in a smaller vessel.

Common mistake
Plaque forms within the wall, beneath the endothelium, not on top of it.

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Consequences: a narrowed artery can cause chest pain, or angina, during exercise. A full block deprives muscle downstream of oxygen, so it dies. If enough muscle is lost, the heart cannot pump effectively. This is a heart attack, or myocardial infarction.

Evaluating data: studies across populations link coronary heart disease with factors such as saturated fat, smoking and high blood pressure. When you evaluate this evidence, look at sample size, spread of data, whether other variables were controlled, and how the data were collected. A strong correlation does not prove a cause.

Watch out: plaque forms within the artery wall, beneath the endothelium. It does not simply stick on top.

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